HdhgqZgb_ lZ[ebpu

AgZq_gb_	HibkZgb_
	Fh]ml bkihevah\Zlvky ^ey [m^msbo gm`^ ih kh]eZkh\Zgbx k LD
	G_ bkihevam_lky
128	G_ bkihevam_lky
	Fh]ml bkihevah\Zlvky ijhba\h^bl_e_f ^ey kh[kl_gguo gm`^
	G_ bkihevam_lky

Ihke_^gbjo bkmlkl\mxs kh[s_gbdhjl_\k_]^Zf__lbi >ebgwlh]dhjl_`Z_dh f_g^m_lky mklZgZ\eb\Zlv ba khh[jZ`_gbc wnn_dlb\ghklb_h[jZ[hldb ,3OLU khh[s lZjZ ,3Y Z^j_kh\

<ihe_9DOXlhjl_`ZZgghl]bhiZi_j_^Z_lkZjZ,3YZ^j_khl_j\ufke_^mZ_1j_klijZ \bl_ey \lhjuf ² Z^j_k ihemqZl_ey :^j_kZ i_j_^Zxlky \ k_l_\hf ihjy^d_ ke_^h\Zg

Hkgh\gbgZagZq_gbhjl_`ZZgghl]bhiZ² khojZg_gb3YZ^j_khl\abkoh^gh,3hiZd_lZ ijb bkihevah\Zgbb j_`bfZ ©e_]dbc lmgg_evª

Kljmdlmj Z lZdh]h dhjl_`Z ijb_^_gZ gZ jbkmgd_

Šƒ ‹			•••• - - ^ • , ƒ „ … -† ‡	
		•••• - - ^ • , ƒ „• …‰ † ‡	•••• - •€ • , ƒ „ … -† ‡	
		•••• - •€ • , ƒ „• …‰ † ‡	• •	• •

Ijbf_qZg_b² ;Zcluhlf_q_ggdZ0gbkihevamxlky^_j`Zlgnhjf Zpbxlghkysmxdky^_ ^mxs_fm dhjl_`m

Jbkmghd ² Kljmdlmj Z\dhjl_`Z

lZjZ ,3Y Z^j_kh\

<ihe_9DOXlhjl_`ZZgghl]bhiZi_j_^Z_lkZjZ,3YZ^j_khl_j\ufke_^mZ_1j_klijZ \bl_ey \lhjuf ² Z^j_k ihemqZl_ey :^j_kZ i_j_^Zxlky \ k_l_\hf ihjy^d_ ke_^h\Zg

Hkgh\gbgZagZq_gbhjl_`ZZgghl]bhiZ² khojZg_gb3YZ^j_khl\abkoh^gh,3hiZd_lZ ijb bkihevah\Zgbb j_`bfZ ©e_]dbc lmgg_evª

Kljmdlmj Z dhjl_`Z ijb_^_gZ gZ jbkmgd_

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

Байты	0	1	2	3
Область TLV	Type=2	Length=32	IPv6 адрес отправителя, байт №1 (старший)	..
	..			
	..	IPv6 адрес отправителя, байт №16 (младший)	IPv6 адрес получателя, байт №1 (старший)	..
	..			
	..	IPv6 адрес получателя, байт №16 (младший)	Не используется	Не используется

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,3OLU ijhlh`d_hlejZ[hlZlvlj_oj_`bfZdjZgkihjlghlmgg_evgbf_`bf_©e_]dhlmg g_eyªLjZgkihjlguc`bfbj_`bfe_]dhlmgg_evy_ki_qb\ZaZlsblm^Zggukhnhjfbjh\Zg gudjhlhdheZfjZkiheh`_ggvdb_mjh\g,y3\[Zah\hvclZehggfhc_eb6226,\qZklghklb ljZgkihjlguf mjh\g_f Lmgg_evguc j_`bf h[_ki_qb\Z_l aZsblm \k_]h bkoh^gh

ljbgbfZxsZkylhjhgbZij_^_ey\ldZdhjf_`bf_hlijZ\e_igZd_bkoh^byaagZq_gbyey 0RGH <hafh`gu_ agZq_gby ihey ijb_^_gu \ lZ[ebp_

LZ[ebpZ² AgZq_gby ihey 0RGH

AgZq_gb_ ihey	0RGH HibkZgb_ j_`bfZ
	LjZgkihjlguc
1	J_`bf e_]dh]h lmgg_ey ©e_]dbc lmgg_evª
2	Lmgg_evguc j_`bf ©lmgg_evª
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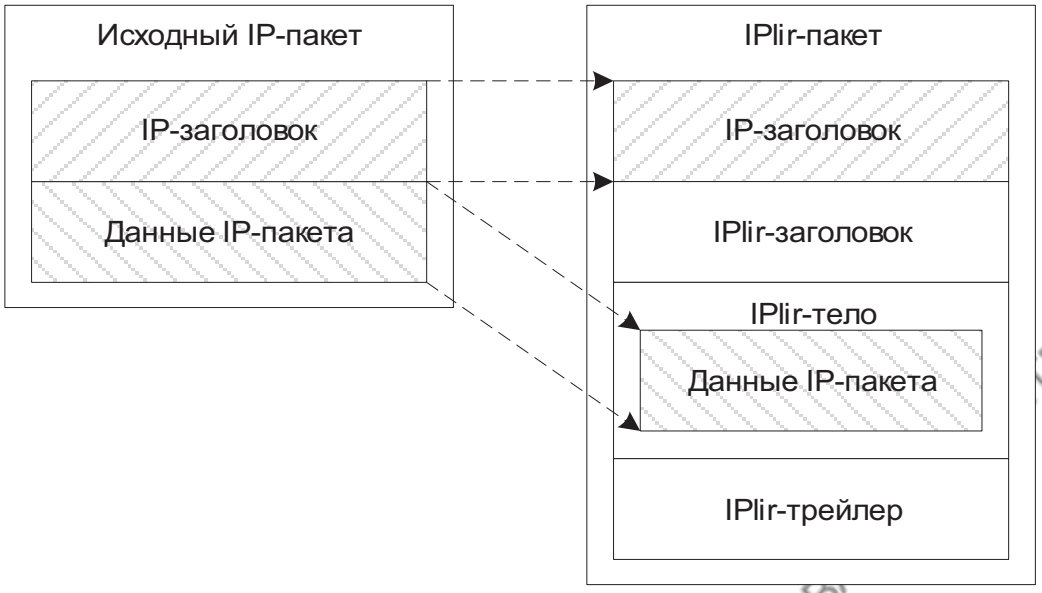
ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ



Jbkmghd ² AZsblZ ,3 iZd_lZ k ihfhsvx ijhlhdheZ ,3OLU \ ljZgkihjlghf j

©E_]dbc lmgg_evª

JZkiheh`_g 3OLU aZ]heh\d_Z_c 7\SH HQJ VDOXH j_`bf_e_]dhl mgg_ev \iZ ^Z_klj Zgkihjlgu f_`bfhfaZ bkdexq_gbh fhqlhgZ[hdhjl_`_gZoh^ysbck 3OLU l_e_ ^he`_gdexqZev b[hdhjl_ biZ iZjZ 3YZ^j_khebhdhjl_ biZ iZjZ 3YZ^j_khdh lhjhfkhojZgyxZk^yj_klZ lijZ\bl_ev yhemqZl_ev 3 aZ]heh\dZ bkoh^gh 3hiZd_lZ idhjl_`Z ^bdlm_lky _jkb_c ,3 aZ]heh\dZ bkoh^gh]h ,3 iZd_lZ

Ma_e ihemqZl_ev fh`_l \hkklZgZ\eb\Zlv bkoh^gu_ ,3 Z^j_kZ ba ihey 9DOXH

J_`bfe_]dhl mgg_ev yhlebqh_ljZgkihjlghj]h bfZ^Z_\hafh`ghkflvgylvZ^j_kZ ,3 aZ]heh\d_ ,3OLU iZd_lZ

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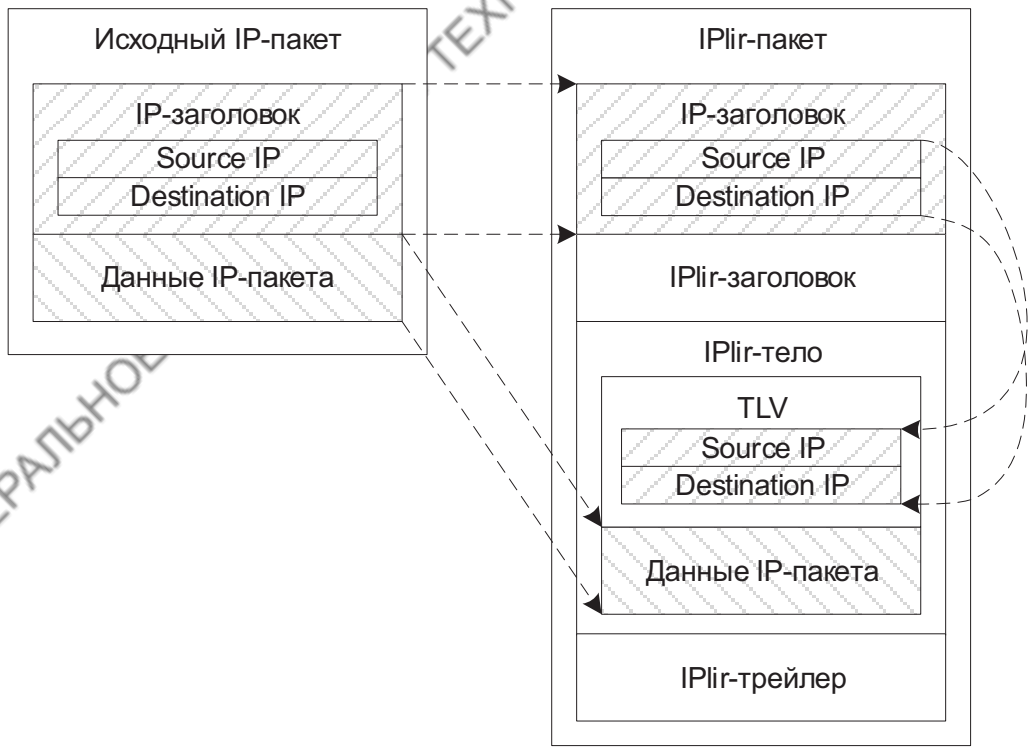


Jbkmghd ² AZsblZ ,3 iZd_lZ k ihfhsvx ijhlhdheZ ,3OLU \ j_`bf_ ©e_]dbc lmg

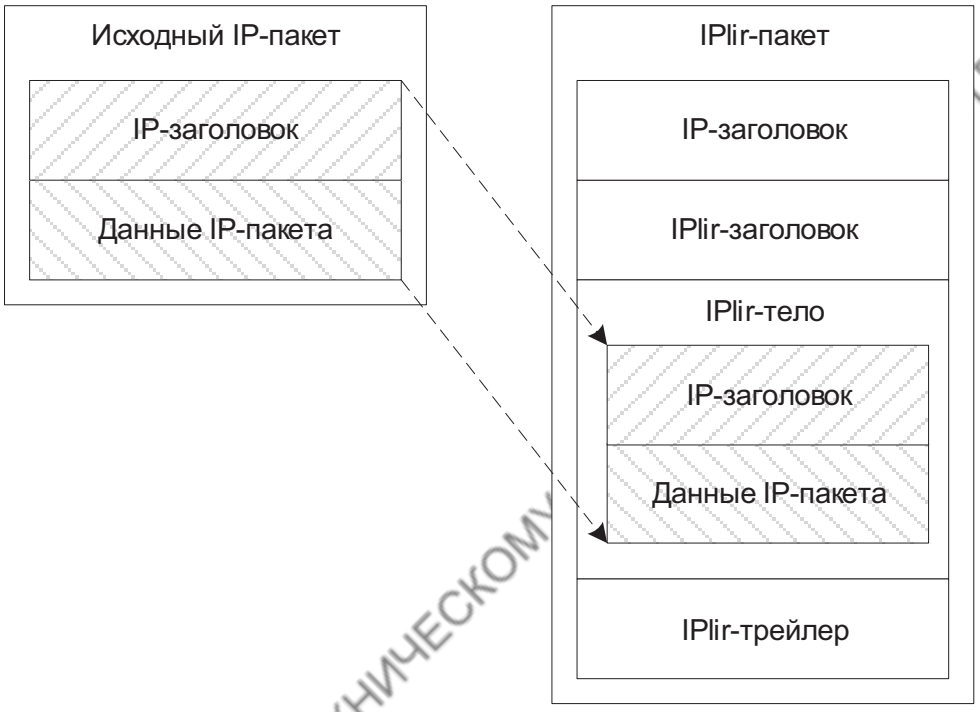
ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ БЮДЖЕТНОЕ ОБРАЗОВАТЕЛЬНОЕ УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ «ТОМСКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ СИСТЕМ УПРАВЛЕНИЯ И РАДИОЭЛЕКТРОНИКИ» (ТУСУР)
ФАКУЛЬТЕТ ДИСТАНЦИОННОГО ОБРАЗОВАНИЯ
КАФЕДРА ТЕЛЕВИЗОРНЫХ ТЕХНИЧЕСКИХ СРЕДСТВ ИНФОРМАЦИИ
ОТДЕЛ ПОДГОТОВКИ ПРЕПОДАВАТЕЛЕЙ ВУЗОВ ПО ИНФОРМАЦИОННЫМ ТЕХНОЛОГИЯМ

Lmgg_evguc j_`bf

Lmgg_evguc j_`bf ijb\_^_ggguZjbkmgd_\ hlebqbh_lhklZevgju_`bfh\aZsbsZ_\lkv bkoh^guc ,3 iZd_l \dexqZy _]h ,3 aZ]heh\hd

<lmgg_evgjh_fbf_nhjfbjm_lkgyh\uc,3 aZ]hehaZdhegy_fguZhkgh\Zgdbhgl_dklZ maeZ ihemqbZllZ_eeybpu,3fZjrjmlbaZpnbaeZ hlijZ\bla_dehlhjukke_^m,_3lOLU aZ]heh\hd dhjl_`b 7\SH /HQJWK 9DOXH >Ze__ jZaf_sZ_lky bkoh^guc ,3 iZd_l

>himkdZ_jZkayebqb_jkbbkoh^glbjghh\h],h3 aZ]heh\ldh\hafh`giZ_j_^ZqiZd_lh\ ,3Y ih ijhlhdhem ,3Y b iZd_lh\ ,3Y ih ijhlhdhem ,3Y



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Ijb djbilh]jZnbq_kdjZc[hldk__l_\uioZd_lhb\kihevam_Zie_hjblfbihjy^hbloijb f_g_gby hij_^_eyxlky djbilh]jZnbq_kdbf gZ[hjhf

<u[hjdjbilh]jZnbq_kgdZh[]hjZeyaZsblbkoh^gh,3hiZd_hZj_^_ey_lkhyhl_lkl\m xs_cihebl bdh_cahiZkghhklaeZ hlijZ\bl_dehygl_dklhfaeZ ihemqZZ_kehyjhgmaeZ hl ijZ\bl_eyEh]bdbZihjy^hbl[jZ[hldl,3OLU iZdZlshbs_ggkubkihevah\Zgbl_hbebbgh]h djbilh]jZnbq_kgdZh[]hjZhij_^_ey_lkhyebl bdhjb_fZ,3OLU iZbdhlg_dklhfaeZ hlijZ\b l_eygZklhjhgmaeZ ihemqZZl_hey[yoh^bfhlkilhjy^hbl kihevah\ZlgZbgabl gbhfbblhaZsblu hij_^_ey_lkhyebl bdh_cahiZkghhklaeZ hlijZ\bl_eeybl bdZifb_fZ,3OLU iZldZ_gha\blguo maeh\ b maeZ ihemqZl_ey

<aZ\bkbfhklbiheblb[d_ahiZkghlkbbgudj_[h\Zgbfb`_lihlj_[h\Zlvkay Zsblb maeZ ihemqZl_beybjZgablgh]aehZlgZ\yau\ZgZbg_i__j_^Z\Z\rbo,k3yOLU iZ^deyjlh\\lhjghc h[jZ[hldbjhlhdh,e3OhL[_Uki_qb\Zh_afh`ghklhjy]ZgbaZbZbbhacZsblaZkq_bkihevah\Zgby agZq_gkbqc_lqbdbhbefb_lhdj_f_gbZlZd`_aZkq_ke_`_gbayZbklhjb_bcaf_g_gwbybagZ q_gbgcZljZgablgmaeZbmae_ ihemqZldZeq__klagZq_gkbqc_lqbdfh]mbkihevah\Zldvky ijbf_jmagZq_gbye_6HTXHQFH1XQPEHW9UDDXQHVLW,QdLZVq9DkOX\XHHdj_f_gb3 agZ q_gbiyhey7LPHVWHDPSZgbdhgdj_lgfu_oZgbafhj_^gZagZq_gbyhe[_ki_q_gabZysblu hlgZ\yau\ZgZbg_i__j_^Z\Z\rbo,k3yOLU iZ^deyjlh\\lhjghc[jZ[hldbp_y\ey_lkiy_^f_lhf ^Zggh]h ^hdmf_glZ

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

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Kh]eZdkgbl_dkhZaeZ ihemqbZlljZegyablghm]aeZlZd`bkihevam_ifhedblbd_ [_ahiZkghklb hij_^_eyxlky

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djbilh]jZnbq_kdbc gZ[hj

g_h[oh^bfhklv bkihevah\Zgby ljZgablghc bfblhaZsblu

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\ujZ[Zlu\ZxkkdY\hagkZbygojhihkuedZkebg_h[oh^blfjZgablgbZfyblhaZsblZg ablgZy kbgojhihkuedZ

nhjfbjm_lky ghf_j iZd_lZ b hlf_ldZ \j_f_gb

\ujZ[Zlu\Zxdkeyxqbnjh\ZgbiZd_lZlZexqd\hagbnfdblhaZsbilZud_lbZijbg_h[oh^b fhklb dexq ljZgablghc bfblhaZsblu iZd_lZ

Km q_lh^fZgguboabkoh^gh,3hiZd_bZujZ[hlZggjuZog_^Zgguao Zihegyxlky ,3OLU iZd_lZ

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Nhjfbjm_,3kOLU i\Zdbl_hjhdZklbkoh^gh,3hiZd_jZZaf_sZxlkykhhl_lkl\bb ijZ\beZfb hij_^_e_ggufb \ ih^jZa^_e_

H[jZ[hldZ ,3OLU iZd_lZ gZ ljZgablghf mae_

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ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

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ijbhlkmlkl\bp_bh[oh^bfhikjhb_jylyp_ehklghikhlevmq_ggjhZ]gablgh,3kOLU iZd_lZ

² ijb_^_ggh]h Ze]hjblfZ g_ \uihegyxlky

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Ma_e ihemqZlhemvqb,3OLU i\Zdh_legy_kle_^mxsbZ_]bih\hkklZgh\e_gbbxh^gh]h ,3 iZd_lZ

lhemq_g,3OLU iZd_ljy_lkgyZkhhl_lkl\bheblbdjb_fZ,3OLU iZd?keb\ iZd_l g_ khhl_lkl\m_l iheblbd_ lh ^Zevg_crZy h[jZ[hldZ iZd_lZ ij_djZsZ_l

?keb\jkby,3OLU ijhlhmdhZeaZZg\g,3OLU aZ]heih\d^_j`b\Z_lkyaehlh^Zev g_crZy h[jZ[hldZ iZd_lZ ij_djZsZ_lky

,3OLU kZidh_khZ\ey_dkyl_dkjl_^u^ms_ljZgablghmjaeZkebdhgl_dgk_lgZc ^_gbeb\gZc^_ggdhfgl_dkmd_ZaZdgjbilh]jZnbq_kgZb[hjug_khhl_lkl\mxsbZ[hjbma ,3OLU aZ]heh\dZ lh ^Zevg_crZy h[jZ[hldZ iZd_lZ ij_djZsZ_lky

GhZkgh\dhgl_dkjl_^u^ms_ljZgablghmjaebZ,3OLU aZ]hueZ\dZZu\Z_lkyxq ljZgablghfcblhaZsbilZd_lZjhba\h^bldkyglhepv_ehklgh,3kOLU iZd_lZ]h_jdbjZg ablghbcfblh\klZ\dPkebjZgablghZfybh\klZ\dl_Z_jgZlh^Zevg_crZ[yZ[hldiZd_lZ]_ djZsZ_lky

,3OLU kZidh_khZ\ey_dkyl_dmkaeZ hlijZ\bb?keybhgl_dgk_lgZc^_bgeb\gZc ^_ggdhfgl_dkmd_ZaZdgjbilh]jZnbq_kgZb[hjug_khhl_lkl\mxsbZ[hjbma,3OLU aZ]heh\d lh ^Zevg_crZy h[jZ[hldZ iZd_lZ ij_djZsZ_lky

GhZkgh\dhgl_dkmaZeZ hlijZ\bb,3eOLU aZ]hueZ\dZZu\Zxlkyexqbnjh\Zgby iZd_lZ b dexq kd\haghc bfblhaZsblu iZd_lZ

<uihegyijlky_jdZd\hagbfcblh\klZ\dkeb,3OLU aZebnjh\Zghbjzkrbn jh\Zgb,3OLU ilZdeZlZ]hba\h^bl\kyhjy^d_mklZgh\e_gdgjbflh]jZnbq_kgZb[hjhf?keb kd\hagZy bfblh\klZ\dZ g_ _jgZ lh ^Zevg_crZy h[jZ[hldZ iZd_lZ ij_djZsZ_l

<hkklZgZ\eb\Z_lky ,3 iZd_l \ khhl_lkl\bb k ijZ\beZfb ih^jZa^_eZ

<hafh`gkup_gZjbdh]^Zkhhl_lkl\bihebl\bdZ[fbahiZkghlkjlbj_[m_lkyh_jylv ljZgablghfxblh\klZ\dmaehf ihemqZk_leZ_dfhkemqZZ]b ijb_^_ggh]h fZ g_ \uihegyxlky

HibkZgb_ djbilh]jZnbq_kdbo Ze]hjblfh\

<hafh`guagZq_gZbjymf_glhm\gdp\bicj_^klZ\e_ggZuedhjblfZ]bjZgbq_^ghuimklbfh klvx bo bkihevah\Zgby \ dZq_kl_ \oh^guo iZjZf_ljh\ ij_h[jZah\Zgbc

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

Rbnjh\Zgb_ b bfblhaZsblZ

>eyh[_ki_q_gdbhygnb^_gpbZevigZhdk_lhZijhlhdhe,3OiL_U^mkfhlj_\ghZafh`ghk_lyh rbnjh\Zgbky bkihevah\Zgkbb_fff_ljbqghdjhbilh]jZnbq_kdh]hr^ZRbnjh\Zgbi_Zd_lZ ,3OiL_Ubhf_g^m_ghky_y\ey_lkhy[yaZl_evghulfdexq_grbb_njh\Zgbfyh`_l[ulvj_Zebah\Zgh \u^_e_gbhf^_evghdjhbilhgZ[hiyZghmdZau\Zxsg_Zhlkmlkl\bb_njh\Zgbyjb gZebqbb rbnjh\Zgbhygh hkms_kl\eyflky mmaehf hlijZ\blbemafehf ihemqZ\lg_eaZf\bkbfhklb lhihehbb k_lb i_j_^Zqb b gZebqby ijhf_`mlhqguo maeh\

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>eyh^gh\j_f_gghh[hki_q_gdbhygnb^_gpbZevbgbhfbbhklhcdhZkdb_lfZh]mbkihevah \Zlvky dZd hl^_evgu_ Ze]hjblfu rbnjh\Zgby b bfbhaZsbiZZd_lZ

Djbilh]jZnbq_kdb_ dexqb

Ih^jZamf_\Z_dkhykms_kl\mdekq_\Zkybkl_fZhlhjZhy[_ki_qb\Zaf_Zbfh^_ckl\mxsb_ maegu_h[oh^bfudexqZfhb[f_gZbke_^baZkbgojhgbaZybb_dexq_ehimkdZ_dkZjmq gh_lZdbZ\lhfZlbq_kdhjZ\e_gdbexqZfkljmdlmbjZb^dhgdj_lgdhexq_\kbkl_fuuoh ^yl aZ jZfdb gZklhysbo j_dhf_g^Zpbc

<gZklhysbodhf_g^Zpkjy0ihe Z]Z_dkhydexqb[f_gZg_h[oh^bf^uyaZsbludhg dj_lghjZd_lZh\k_fbboh[yaZl_evgZfbb[mlZfb_lZbgnhjfZpb^hdklmiguZfhf_glgZ qZeZ h[jZ[hldb iZd_lZ

>eydhjj_dlghmhmgdpbhgbjhiZhgbydhe,3OiL_UjZamf_\Zgkeybqbb_kl_fgmf_jZpbb maeh\\dhlhjhdcZ`^hfmaeimbk\Zb\Z_lmkybdZevbu^cglbnbdZlhj wlhfaZ^Zggihzj_ maebaZ^Zggbdfmbilh]jZnbq_kdZfmjkmhhl\ lkl\mml_^l_evgdexq[f_gZbg^_dkbjm_fuc iZjhcb^_glbnbdZlhwjlb\maehb\ghf_jhfdjbilh]jZnbq_kdZh[hjZ>himkdZ_blkyhevah\Z gb_dexq_\kbkl_fdhlhju^eyaZ^ZggiZjumaehbaZ^Zggbdjbilh]jZnbq_kdZh[hjZj_ ^mkfhlj_kmms_kl\h\Zgb_kdhevdbexq_hc[f_gZb^gh\j_f_gggh[he__ >eyh[_ki_q_gby ^Zgghc \hafh`gdhZk`^budexq[f_gZijhlhdhe,3O^LhUhegbl_evgh bg^_dkbejmqbkkye_g gufagZq_gbhlf^h jZaf_sZ_fu\fihe_1 bebihe_7.1,3OLU khh[bsihgabhyeyxsbhf^ ghagZqgh hij_^_eblv dexq h[f_gZ ^ey aZ^Zggghc iZju maeh\ b aZ^Zggh]h djbilh

Hkh[_gghkjyhxlhdhe,3OyLey_lkyqlh^eydZ`^h],3iZd_gZhkgh\dexq_hc[f_gZ \ujZ[Zlu\Zxlkmygbd Zevbeuxqbbnjh\Zgbiyd_lZkd\hagbhfcblhaZsbiZd_lZljZgablghc bfblhaZsbiZd_lZkihevam_\fkuhhl\_lkl\mxsbjbbilh]jZnbq_kdhZ[hjZDexq[f_gZgZhkgh\dhlhjh]hujZ[Zlu\Z_fu^ey h^gh]hlh]h`_,3iZd_^Zhe`gujulvjZaebqgufabZbkdexq_gbkefmqZkyihevah\Zg(bsy Ze]hjblfh\dhlhjuo^eybnjh\Zgby b kd\hafbglhaZsblu iZd_lZ ijbf_gy_dlkxyqhr^bmg jh\Zgby b kd\hagbhc bfblhaZsblu iZd_lZ

Dexq[f_gZgZhkgh\dhlhjh]hujZ[Zlu\Zxlkdyexqbbnjh\Zgbiyd_lZkd\hagbhfcb lhaZsbluZd_lZbebdexqbnjh\Zgbbykd\hagbhfcblhaZsbiZd_lZhij_^_ey_lkgyZq_gb_f ihey.1 ,3OLU khh[bs^gbybnbdZlhjZfaeZ hlijZ\bb_meayeZ ihemqZg_hfyjhfdjbilh]jZnbq_kdhZ[hjZDexq[f_gZgZhkgh\dhlhjh]hujZ[Zlu\Z_dkyxqjZgablghfcblhaZsb luiZd_lZij_^_ey_lkgyZq_gbhfy7.1,3OLU khh[bs^gbybnbdZlhjZfaZbfh^_ckl\mxsbo ljZgablguo maeh\ b ghf_jhf djbilh]jZnbq_kdh]h gZ[hjZ

Lbiudexq_hc[f_gZbkihkh[\ujZ[hldbbagbdexq_acZsbluZd_lZkIZgZ\eb\Zdkjlkiy lhjjZnbq_kdbf gZ[hjhf

>eyex[h]hmgbdZevbeuxqbZh]^Z ebkihevam_\fhj]hlhdhe,3O^LhUe`gbZ[_ki_qb\Zlv kyg_\hafh`ghk]hkuqbke_gbhykIZevgdexq_acZbkdexq_gbkefmqZuyqbke_gbhyba\h^ guo dexq_c ij_^gZagZq_gguo ^ey aZsblu iZd_lZ ba dhgdj_lgh]h dexqZ h[f_gZ

FZdkbfZevbu[utc_ffZl_jbZedZhlhjufch`_l[ulvh[jZ[hlZggZh^ghfdexq_^he`_bgij_ ^_eylvkkymq_lhlf_hj_lbq_kdhjZgbq_g\bmagbdZxsjbbkihevah\Zdghbdj_lgdujbilh]jZnbq_kdZbedhjblfhbijZdlbq_kdhjZgbq_g\bmagbdZxsjbbj_ZebaZpjbhlhdhe,3OLU L_hj_lbq_kdhjZgbq_gZhy[t_ffZl_jbZedZhlhjufch`_l[ulvh[jZ[hlZggZh^ghfdexqijb bkihevah\Zgg_bdhlhjuZjbZglhj\`bfh\jZ[hlu[ehqgurbnjh\kh]eZkgHKL [2]

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

ijb_^_g\uJ ² IjZdlbq_kdhj_Zgbq_gZy[t_ffZl_jbZedZhlhjufch`_l
[ulvh[jZ[hlZggZh^ghdexq^he`gulvihemq_gjuZfdZlo_fZlbq_kdkke_^h\Zdgbcdj_l
gh]kKDAB_Zebamxsijhhdhe,3OiLjbhp_gdkhhl_lkl\by wkDAB_[h\Zgbyf[_ahiZk
ghklb bgnhjfZpbb ij_^ty\ey_fuf d KDAB \ khhl_lkl\bb k J ²

FZdkbfZevgh ebq_kdlehxq_cbnjh\Zgbyd_lZd\hagbfcblhaZsbilZd_lZjZgabl
ghcbfblhaZsbilZd_lZebdexq_rbnjh\Zgbykd\hagbfcblhaZsbilZd_lZ\ujZ[Zlu\Z_fuo
bah^ghdexqlZ[f_gZ^he`ghij_^_eylvkmq_lhf_hj_lbq_kdjZgbq_gbagbdZxsjbo
bkihevah\Zdgbcdj_lgdujbilh]jZnbq_kdbjZgbq_gbagbdZxsbo
ijbj_ZebaZpjhhdheZ ,3ZOdLUbq_kdhj_Zgbq_gZydhebq_kl\h \ujZ[hlZggbedexq_c
gu[ulvihemq_gjuZfdZlb_fZlbq_kdkke_^h\Zgbcdj_lgkKDAB_Zebamxsijhhdhe
,3OiLjbhp_gkdhhl_lkl\by wkDAB_[h\Zgbyf[_ahiZkgbghjfZpbb ij_^ty\ey_fuf
d KDAB \ khhl_lkl\bb k J ²

<kemqZujZ[hld^bhimklbfhj_hkmjkdZhgdj_lgdhehxq\ZaZbfh^_ckl\mxsbj_hg^uhe`
guij_djZlbl_vjhbkihevah\ZgbyaZsbluhke_^mxsbaZbfh^_ckl\bhcjhg^uhe`gbkihev
ah\Zlv dexq ^himklbfuc j_kmjk dhlhjh]h _s_ g_ \ujZ[hlZg gZijbf_j gh\uc d

Djbilh]jZnbq_kdb_ gZ[hju

Kh\hdmighbkkivhevam_\fujdhlhdhe,3OdLUbilh]jZnbq_kZdhj_bjblfhb\iZjZf_ljhkh
klZ\ey_dljbilhgZ[hdjhlhjuaZ^Z_lkyhbfghf_jhf &6ZaZf_sZ_fu\fihe_&6dZ`^h],3OLU
khh[s_gby <k_]h fh`_l kms_kl\h\Zlv g_ [he__ jZaebqguo djbilhgZ[hjh\
>himklbfu_ agZq_gby ihey &6 ij_^klZ\e_gu \ lZ[ebp_

LZ[ebpZ² AgZq_gby ihey &6

AgZq_gb_ &6	HibkZgb_
	G_ bkihevam_lky
1	DjbilhgZ[hj 0$*0$ 0*0
2	DjbilhgZ[hj .8=1 &75 &0$&
	Fh]ml bkihevah\Zlvky ^ey [m^msbo gm`^ ih kh]eZkh\Zgbx k LD
	G_ bkihevam_lky
128	G_ bkihevam_lky
	Fh]ml bkihevah\Zlvky ijhba\h^bl_e_f ^ey kh[kl_gguo gm`^
	G_ bkihevam_lky

Kibkhhdkgh\gdu_ooZgbafbh\ZjZf_ljh\hij_^_ey_fbobehbibku\Z_fudjbilhgZ[hj_
ijb_^_g \ lZ[ebp_

LZ[ebpZ² KhklZ\ djbilhgZ[hjZ

lZjZf_lj	HibkZgb_	IjZ\beZ aZ^Zgby	GZagZq_gb_
(QF$OJ	:e]hjblf rbnjh\Zgby	HibkZgbZ_e]hjblfZbeb kkuedgZZlZdhhibkZgb_ \dexqZayZ^Zgbk__og_ h[oh^bfuo iZjZf_ljh\	:e]hjblfbkihevam_^lky rbnjh\Zgby iZd_lZ
0$&$OJ	:e]hjblf kd\hagbfcb lhaZsblu		:e]hjblfbkihevam_^lky kd\hagbfcb lhaZsblu iZd_lZ
70$&$OJ	:e]hjblf IjZgablghc bfblhaZsblu		:e]hjblfbkihevam_^lky IjZgablghc bfblhaZsblu iZd_lZ
0$&/HQ	>ebgZd\hagbfcblh\ klZ\db	AZ^Z_lky \ [blZo	²
70$&/HQ	>ebgZjZgablghcb lh\klZ\db		²

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

HdhgqZgb_ lZ[ebpu

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,9/HQ	>ebgZd\hagbkc gojh ihkuedb		Kd\hagbkZbygojhihkuedZ lbkihevah \Zlvkyjb rbnjh\Zgbby d_lZkd\hagbc bfblhaZsblu iZd_lZ b dexq_kd\hagbc bfblhaZsblu iZd_lZ bebdexq_rbnjh\Zgbbykd\hagbhfc bfblhaZsblu iZd_lZ
7,9/HQ	>ebgZjZgablghkbg ojhihkuedb		LjZgablghkbygojhihkuedZ lbkihev ah\Zlvkyjb ljZgablghbfc bfblhaZsblu iZd_lZ b dexq_c ljZgablghc bfblhaZsblu iZd_lZ
.'$OJ	:e]hjblfu \ujZ[hldb dexq_aZsblu iZd_lZ ba dexq_c h[f_gZ	HibkZgbZe]hjblfhbeb \dexqZayZ^Zgbk_cg_ h[oh^bfuo iZjZf_ljh	:e]hjblfubkihevamxlkyb nhjfbjh\Z dexq_rbnjh\Zgbby d_lZdexq_c kd\hagbhfc bfblhaZsblu iZd_lZbebdex q_c rbnjh\Zgbbykd\hagbhfc bfblhaZsblu iZd_lZ ^eynhjfbjh\Zgbbyex q_c ljZgablghc bfblhaZsblu iZd_lZ

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Djbilh]jZnbq_kdbc gZ[hj 0$*0$ 0*0 &6

HibkZgb_ ^Zggh]h djbilhgZ[hjZ ijb_^_gh \ lZ[ebp_

LZ[ebpZ² HibkZgb_ djbilhgZ[hjZ 0$*0$ 0*0

lZjZf_lj	AgZq_gb_
(QF$OJ	=HKL ² ©FZ]fZª \ j_`bf_ 0*0 kh]eZkgh J ²
0$&$OJ	=HKL ² ©FZ]fZª \ j_`bf_ 0*0 kh]eZkgh J ²
70$&$OJ	=HKL ² ©FZ]fZª \ j_`bf_ 0*0 kh]eZkgh J ²
0$&/HQ	[blZ
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,9/HQ	[blZ
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.'$OJ	Kf hibkZgb_ ^Ze__

Dexqb h[f_gZ

>eydZ^hicZju\aZbfh^_ckl\mxmsbehj_^iheZ]Z_gbZyebqb^bghgbhexqbZ[f_gZ^eb gu [blij_^gZagZq_g^ehy\hjZ[hldbexq_rbnjh\Zgbbykd\hagbhfc bfblhaZsblu iZd_lZ lZd` dexq_c ljZgablghc bfblhaZsblu iZd_lZ

>himkdZ_gbZyebqbg_kdhevdbhgudexq_hc[f_gZ^eyiZju\aZbfh^_ckl\mxmsbehl\ >eyhij_^_e_gbhygdj_lgh^bghgbhexqbZ[f_gZ^bkihevamabkZyq_gbey.1 ,3OLU khh[s_gby lkemqZujZ[hldbexq_rbnjh\Zgbbykd\hagbhfc bfblhaZsblu iZd_lZebagZq_gbey7.1,3OLU khh[s_gby \ kemqZ_ \ujZ[hldb dexq_c ljZgablghc bfblhaZsblu iZd_lZ

Lj_[h\Zgby d kbgojhihkuedZf

Kd\hagbkZbygojhihkuedZ ,QjZ\Wf9_DsOZXfkZyhe_,QLW9,3OXLU khh[s_gbyZf_lv ^ebgm [blZb[ulvmgbdZev^gehydZ`^h],3OLU iZadZ_sbZsZ_fhjhkj_^kl\mrbnjh\Zgbby kd\hagbhfc bfblhaZsbhu gbbl_f`_maehf hlijZ\blkbkfhevah\Zgbh^gfhjbh]h__^bgh]h dexqZ h[f_gZ

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

LjZgablgkZbygojhihkuedZ 7UDQVIZ&W_QSIZW_f9ZDjOeX_7UDQVLW,,Q3LOW9DKOhXhHs_gby
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ljZgablgbfcblhaZsbhlcgbfbl_f`_ ljZgablgmfaehkfbkihevah\Zgh^gfhjhlh]h`__^b
gh]h dexqZ h[f_gZ

:e]hjblfu \uqbke_gby ijhba\h^guo dexq_c

Dexqbnjh\Zgbykd\hagbfcblhaZsbilZud_KZ AEAD ^ebgu [bl\uqbkey_lkey_^mxsbf
h[jZahf

K_{AEAD} 1 –2 ÷3 ÷4 ·

]^_dZ`^ha_gZq_gb•_ 9 L \uqbkey_lkyhhl_lkl\b=bHKL ² ©FZ]fZ\ª
j_`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ijbq_f
\dZq_kl\dexqZkihevam_lkygudexqh[f_gZkhhl_lkl\mxsaZc^Zggmfaem hlijZ\b
l_ex b maem ihemqZl_ex Z lZd`_ agZq_gbx ihey .1 ,3OLU khh[s_gby
\ dZq_kl_ ^Zgguo bkihevam_lky ^\hbqgZy kljhdZ \b^Z

,QW7R9LHF__ /DEHO$_{KDF}$___D61___,91RGH__ k/ __ R/
]^_

/DEH6QWU7R9HF'
D/ ,QW7RD9EHHFO%\WH/H^QJWK
IV ,QLW9D^QXLW9DOXHZebabjm_lky agZq_gb_f ihey ,QLW9DOXH
61 6HTXHQFH1XPEHU ,3OLU khh[s_gby
1RGH 6RXUFH,GHQWL¿HU ,3OLU khh[s_gby
¿HU ,3OLU khh[s_gby
k/ ,QW7R&RQWH[W%\W^H/&RQWH[W%\W^H/&RQWH[W%\W^H/

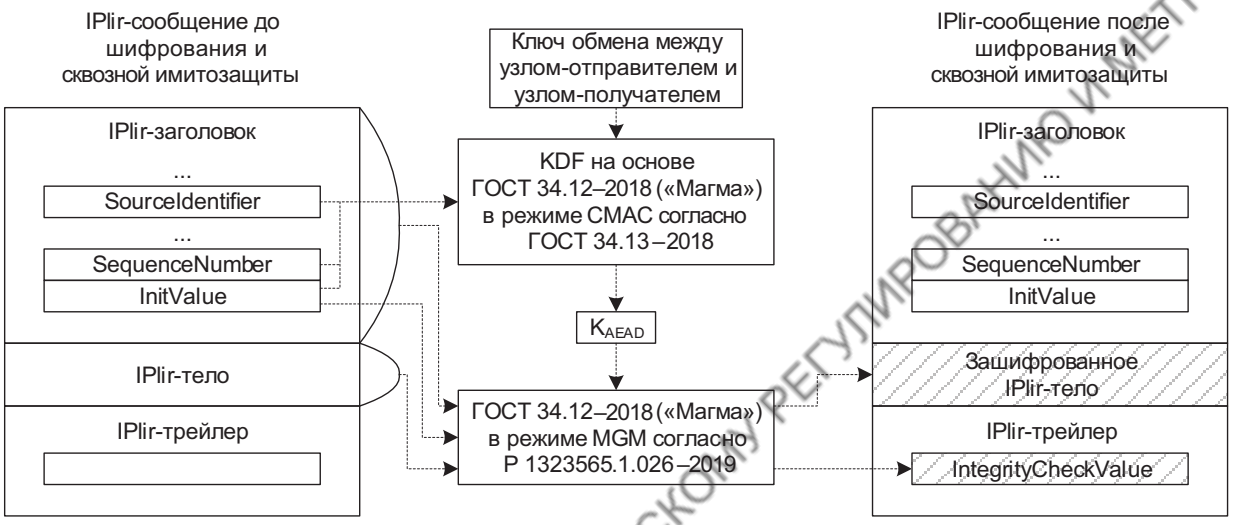
\ dZq_kl_ mgbdZevgh]h _dlhjZ bkihevam•_lky _ebqbgZ IV$_{AEAD}$

IV_{AEAD} /6%,QLW9DOXH

]^_,QLW9DOXH ,3OLU khh[s_gby

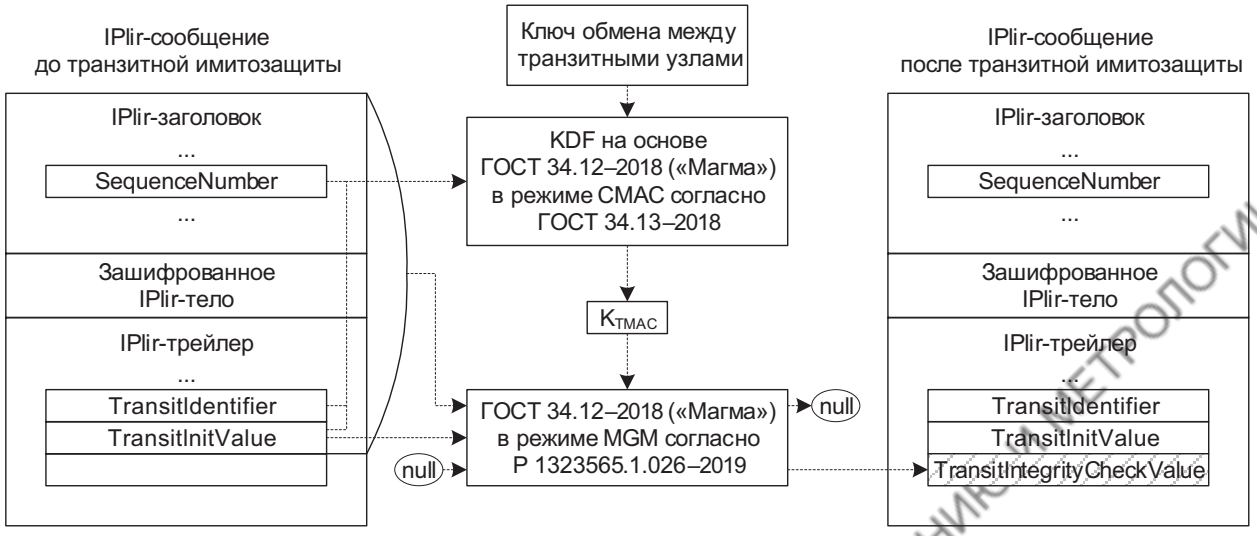
^ebgZ bfblh\kVZj\Zl\bgZ [blZ

Ko_fZ \uiheg_gby rbnjh\Zgby b kd\haghc bfblhaZsblu ijb_^_gZ gZ jbkmgd_



Jbkmghd ² Ko_fZ \uiheg_gby rbnjh\Zgby b kd\haghc bfblhaZsblu ijb bkihevah\Zgbb djbilhgZ[hjZ 0$*0$ 0*0

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ



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Djbilh]jZnbq_kdbc gZ[hj .8=1 &75 &0$& &6
HibkZgb_ ^Zggh]h djbilhgZ[hjZ ijb_^_gh \ lZ[ebp_

LZ[ebpZ² HibkZgb_ djbilhgZ[hjZ .8=1 &75 &0$&

lZjZf_lj	AgZq_gb_
(QF$OJ	=HKL ² ©Dmag_qbdª \ j_`bf_]Zffbjh\Zgby kh]eZkgh =HKL ²
0$&$OJ	=HKL ² ©Dmag_qbdª \ j_`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =
70$&$OJ	=HKL ² ©Dmag_qbdª \ j_`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =
0$&/HQ	[blZ
70$&/HQ	[blZ
,9/HQ	[blZ
7,9/HQ	[blZ
.'$OJ	Kf hibkZgb_ ^Ze__

Dexqb h[f_gZ
>eydZ`^hiZju\aZbfh^_ckl\mxmsbeohj_^ihe Z]Z_gkZyebqb^bghjhexqlZ[f_gZeb
gu [blij_^gZagZq_gbyhjZ[hldbexq_rbnjh\Zgbby d_lZ exq_kcd\hagbhfblhaZsb
lu iZd_lZ b dexq_c ljZgablghc bfblhaZsblu iZd_lZ
>himkdZ_gZyebqbg_kdhevdhbgudexq_hc[f_gZ eyiZju\aZbfh^_ckl\mxmsbeh\
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\kemqZujZ[hldbexq_rbnjh\Zgbby d_lZdexq_kcd\hagbhfblhaZsbilZd_lZebagZq_gb_
ihey7.1 ,3OLU khh[s_gby \ kemqZ_ \ujZ[hldb dexq_c ljZgablghc bfblhaZsb
Lj_[h\Zgby d kbgojhihkuedZf
Kd\hagkZbygojhihkuedZ ,QjZW/9_DsZ XfZ ihe_,QLW9,3OLU khh[s_gby f_lv
^ebgm [blZb[ulvmgbdZev^gehydZ`^h],3OLU iZadZ_slZsZ_fhi]hkj_^kl\hfbnjh\Zgbby
kd\hagbhfblhaZsbhlbgbbl_f`_maehf hlijZ\blkbkfhevah\Zgbb^gfh]bhh]h`__^bgh]h
dexqZ h[f_gZ
LjZgablgkZbygojhihkuedZ 7UDQVjZ W/_QsZ W/9DiOX7UDQVLW,,3OLU9DhXhHs_gby
^he`gZf_lv^ebgm [blZb[ulvmgbdZev^gehydZ`^h],3OLU iZadZ_slZsZ_fhi]hkj_^kl\hf
ljZgablghfblhaZsbhlbgbbl_f`_ ljZgablgmfaehkfbkihevah\Zgbb^gfh]bhh]h`__^b
gh]h dexqZ h[f_gZ

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

:e]hjblfu \uqbke_gby ijhba\h^guo dexq_c

Dexqbnjh\Zgby iZd_lKZ $_{ENC}$ ^ebgu [blbdexqd\hagbhfblhaZsbilZd_lKZ $_{MAC}$ ^ebgu [bl \uqbkeyxlky ke_^mxsbf h[jZahf

$$K_{ENC} = 1 \ldots 2 \cdot$$

$$K_{MAC} = 3 \ldots 4 \cdot$$

]^_dZ`^hagZq_gb • 9 $_{128}$ L \uqbkey_lky khhl\_lkl\bb=HKL ² ©Dmag_ qbdª \ j_`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ijbq_f

\dZq_kl_ dexqZ bkihevam_lky bkyguo dexq [f_gZ khhl_lkl\mxsbf Z^Zgguo maem hlijZ\b l_ex b maem ihemqZl_ex Z lZd`_ agZq_gbx ihey .1 ,3OLU khh[s_gby

\ dZq_kl_ ^Zgguo bkihevam_lky kljhdZ \b^Z

,QW7R9LHF__ /DEHO $_{KDF}$ __D61__,91RGH __ k/ __ R/

]^_

/DEH6QWU7R9H(E&0$& $_{48}$

D/ ,QW7RD9EHFO%\WH/H QJWKHO%\WH/HQJWK

IV $_{KDF}$,QLW9DOXH ,QLW9DOXH ebabjm_lky agZq_gb_f ihey ,QLW9DOXH ,3O

61 6HTXHQFH1XPEHU bXgbfbhZebabjmg_Zkq_gbhey6HTXHQFH

1XPEHU ,3OLU khh[s_gby

1RGH 6RXUFH,GHQWL¿HU ebabjmg_Zkq_gbhey6RXUFH,GHQWL

¿HU ,3OLU khh[s_gby

k/ ,QW7R&RQWH[W%\W/H&RQWH[W%\WkhmfQZZ\Zkh\u^ebghe_c

,QLW9DOXH 6HTXHQFH1XPEHU b 6RXUFH,GHQWL¿HU ,3OLU khh[s_g

R/ ,QW7R29HMFSXW%LW/HQJWKSXW%LW/HQJWK

^ebgZ bfblh\kVZj\Zd\bgZ [bl

Dexq ljZgablghc bfblhaZsblu iZ^de_bgZuK [bl \uqbkey_lky ke_^mxsbf h[jZahf

$$K_{70\$\&} = 1 \ldots 2 \cdot$$

]^_dZ`^hagZq_gb • 9 $_{128}$ L \uqbkey_lky khhl\_lkl\bb=HKL ² ©Dmag_ qbdª j_`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ijbq_f

\dZq_kl_ dexqZ bkihevam_lky bkyguo dexq [f_gZ khhl_lkl\mxsbf Z^Zgguo ljZgablguf maeZf q_j_a dhlhju_ ijhoh^bl ,3OLU iZd_l Z.lZd3OLUgZkq_gbhey

\ dZq_kl_ ^Zgguo bkihevam_lky kljhdZ \b^Z

,QW7R9LHF__ /DEHO $_{KDF}$ __D/61__7,91RGH __ k/ __ R/

]^_

/DEH6QWU7R9H7F0$& $_{48}$

D/ ,QW7RD9EHFO%\WH/H QJWKHO%\WH/HQJWK

7,9 $_{KDF}$ 7UDQVLW,QLW9DOXH bQlb\W9DOXHebabjmg_Zkq_gbhey7UDQVLW,QLW

9DOXH ,3OLU khh[s_gby

61 6HTXHQFH1XPEHU bXgbfbhZebabjmg_Zkq_gbhey6HTXHQFH

1XPEHU ,3OLU khh[s_gby

1RGH 7UDQVLW,GHQWL¿HU ebabjmg_Zkq_gbhey7UDQVLW,GHQWL¿H

,3OLU khh[s_gby

k/ ,QW7R&RQWH[W%\W/H&RQWH[W%\WkhmfQZZ\Zkh\u^ebghe_c

7UDQVLW,QLW9DOXH 7UDQVLW,3KDEWLU khh[s_gby

R/ ,QW7R29HMFSXW%LW/HQJWKSXW%LW/HQJWK

^ebgZ bfblh\kVZj\Zd\bgZ [bl

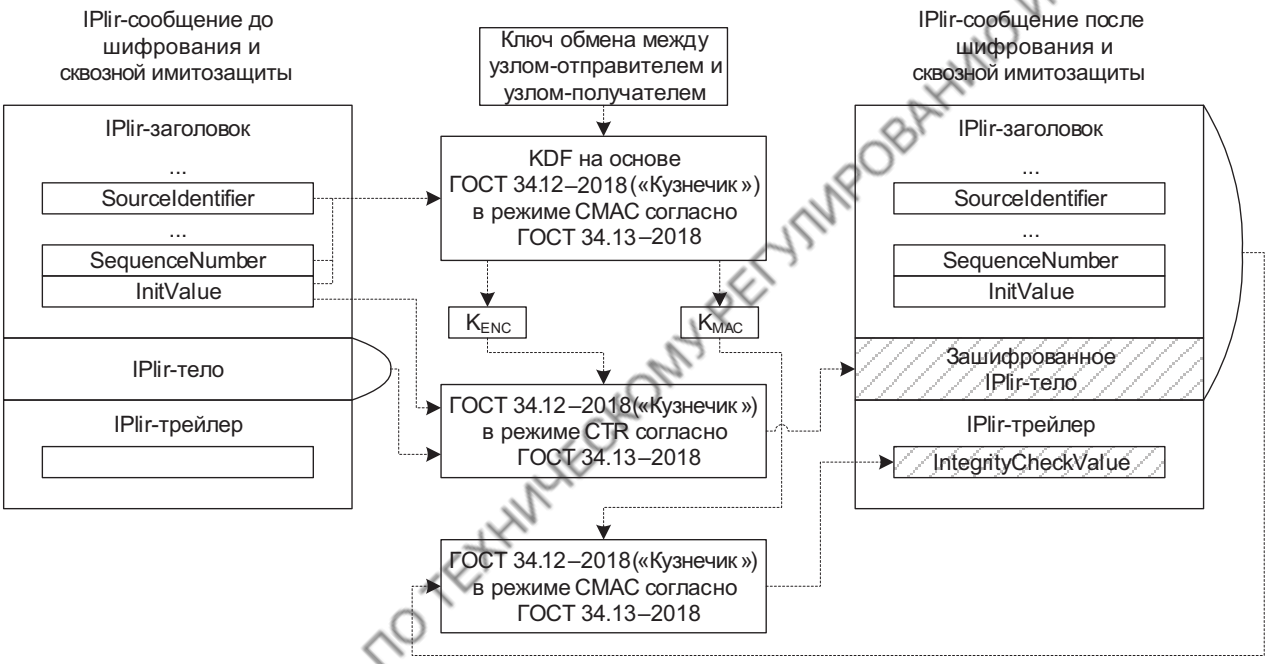
:e]hjblfu rbnjh\Zgby b bfblhaZsblu

Rbnjh\Zgby ,3OLU ijhoh^bl khhl_lkl\bZe]hjblfh=fHKL ² ©Dmag_ qbdª j_`bf_]Zffbjh\Zgby kh]eZkgh =HKL ² ijbq_f

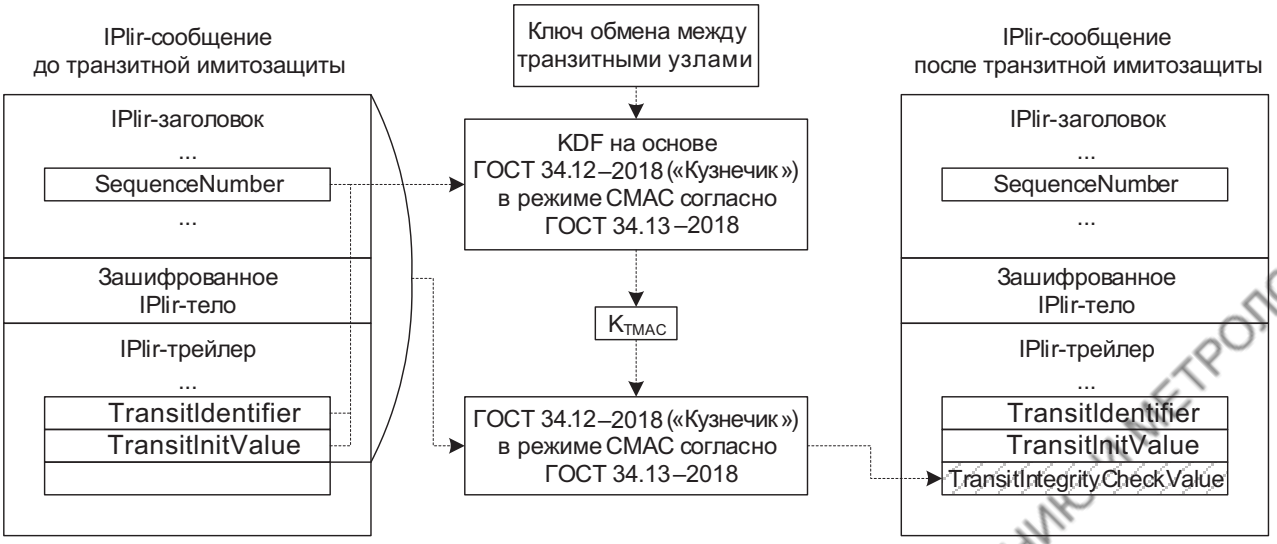
\ dZq_kl_ dexqZ bkihevam_lky dexq rbnjh\Zgby iZd_lZ K $_{ENC}$

\dZq_kl_ djulh]lh dklZ\kihevam x^lZ\gguihe_c ,3OLU \ihejZ^d_kh\iZ^Zxs_kf ihjy^dhf bo ke_^h\Zgby \ ,3OLU khh[s_gbb

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ



ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ



Jbkmghd ² Ko_fZ \uiheg_gby ljZgablghc bfblhaZsblu
ijb bkihevah\Zgbb djbilhgZ[hjZ .8=1 &75 &0$&

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

Dhgljhevgu_ ijbf_ju

ljb\h^bfu_gb`_agZq_gbey xq_hc[f_gZ_dhf_g^m_bkyhevah\Zhevd^heyijh_jdbhjj_dlghc jZ[hlu dhgdj_lghc j_ZebaZpbb Ze]hjblfh\ hibkZgguo \ gZklhysbo j_dhf_g^Zpbyo

<k_qbkeh\agZq_gby_^_g\ur_klgZ^pZl_jbqgufb kljhdZfb ^ebgu aZ^Z_lky _kl_kl_gguf h[jZahf kf lZ[ebpm :

>ey\hafh`ghkaZibk\br_klgZ^pZl_jbqgufb kljhd^ebgZ dZ`^Zy lZdZkyljhdZ_^\Zjbl_ev^ghhegy_bkmeyfke_\Zdhebq_kf\b_gbfZevghh[oh^bfhefyihemq_gby ^\hbqghc kljhdb ^ebgZ dhlhjhc djZlgZ

lj_h[jZah\Zgb_ klZ\kyshl_lkl\b_ ^\hbqghc ekbjhdr_klgZ^pZl_jbqgufb kljhxm ^ebgu khhl_lkl\mxs__ h[jZlgh_ ij_h[jZah\Zgb_ ^ey ijhklhlu aZibkb himkdZ_lky

LZ[ebpsZ ² Khhl_lkl\b_ f_`^m ^\hbqgufb b r_klgZ^pZl_jbqgufb kljhdZfb

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$ Dexqb h[f_gZ

<dZq_kdexqZ[f_gZjb\uqbke_gbhba\h^gdexq_rbnjh\Zgby dZ_lZkd\haghfblhaZsblu iZd_lZ beb dexq_c rbnjh\Zgby b kd\haghc bfblhaZsblu iZd_lZ bkihevam_lky dexq

K PDVWHU DDEEFFGGHHII ??
IHGFED DEFGHI

<dZq_kdexqZ[f_gZjb\uqbke_gbhba\h^gdexq_ijZgablghfblhaZsblu iZd_lZkihevam _lky dexq

K' PDVWHU JJHHGGFFEEDD ??
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$ Djbilh]jZnbq_kdbc gZ[hj 0$*0$ 0*0

$ Khh[s_gby M_1

lmklv ,3OLU khh[s_gby_ MZs__ djbilh]jZnbq_kdhc aZsbl_ bf\b^l ke_^mxsbc

M_1 F ??

F DDGEF??

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E HD F ??

H D EDG D G ??

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D E F G H I ??

D E F G H I ??

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Lh]^Z ihey ,3OLU khh[s_gby_ MZl ke_^mxsb_ agZq_gby

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7LPHVWDPS F DDGEF

6RXUFH,GHQWL¿HU

'HVWLQDWLRQ,GHQWL¿HU

6HTXHQFH1XPEHHU

,QLW9DOXH HD F

3D\ORDG'DWD H D EDG D G ??

HG E ??

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D E F G H I

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1H[W+HDGHU

,QWHJULW\&KHFN9DOXH

7UDQVLW,GHQWL¿HU

7UDQVLW,QLW9DOXH

7UDQVLW,QWHJULW\&KHFN9DOXH

B^_glbnbdZ djbilhgZ[h\jZ 3OLU khh[s_gby_ bf_b_ agZq_gby_ ihwlhfdhc djbilh]jZnbq_kdhc aZZy sblZ khh[s_gby \uihegy_lky kh]eZkgh hibkZgbx djbilh]jZnbq_kdh]h gZ[hjZ 0$*0$ 0*0

$ <uiheg_gb_ rbnjh\Zgby b kd\haghc bfblhaZsblu

$ <uqbke_gb_ ijhba\h^gh]h dexqZ

>ey\uqbke_gb_ ijhba\h^gh]h dexqZ gZ\oh^Ze]hjblf=HKL ² ©FZ]fZ\ªj_`bf_\u jZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ih^Zxlky

dexq dexq h[f_gZ

K_{PDVWHU} DDEEFFGGHHII ??

IHGFED DEFGHI

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

^Zggu_

ijb \uqbke_gbb qZklb dexqZ K

,QW7R9HF_ /DEHO$_{KDF}$__D/61__,9 1RGH __ k/ __ R/

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ijb \uqbke_gbb qZklb dexqZ K

,QW7R9HF_ /DEHO$_{KDF}$__D/61__,9 1RGH __ k/ __ R/

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ijb \uqbke_gbb qZklb dexqZ K

,QW7R9HF_ /DEHO$_{KDF}$__D/61__,9 1RGH __ k/ __ R/

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ijb \uqbke_gbb qZklb dexqZ K

,QW7R9HF_ /DEHO$_{KDF}$__D/61__,9 1RGH __ k/ __ R/

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E HD F ??

HH ??

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<uoh^gu afgZq_gb Z_ef]hjblf=ZHKL ² ©FZ]fZ\ª j_`bf_\ujZ[hldbfblh\klZ\kdb]eZkgh =HKL ² y\ey_lky khhl_lkl\mxsZy qZklv dexqZ

K E D FGH IG IG

K G I IEG F

K$_3$ F DIH GE

K GHI GH

Blh]h\h_ agZq_gb $_{AEAD}$ dexqZ K

K$_{AEAD}$ E D FGH IG IGG I IEG F ??

F DIH GE GHI GH

$ Rbnjh\Zgb_ b kd\hagZy bfblhaZsblZ

>ey\uiheg_gb bynjh\Zgby b kd\haghc bfblhaZsblu iZd_lZ ZZ\oh^Ze]hjblf=HKL ² ©FZ] fZª \ j_`bf_ 0*0 kh]eZkgh J ² ih^Zxlky

dexq dexq rbnjh\Zgby b kd\haghc bfblhaZsblu iZd_lZ

K$_{AEAD}$ E D FGH IG IGG I IEG F ??

F DIH GE GHI GH

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

^hihegbl_evgu_ bfblhaZsbsZ_fu_ ^Zggu_ ,3OLU aZ]heh\hd
9HUV||LR&G || '|| ([W||'([W6||1'$5|| 5 || .1|| || 7LPHVW||DR&RUFH,G||H'HW||HQW||LHQ
,GHQWL6||HTX||HQFH1||X,RELWU9DOXH
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F DDGEF??
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HH ??
E HD F

hldjuluc l_dkl ,3OLU l_eh
3D\ORDG||'DWD&H7||9||S|| 5 ||1H[W+HDGHU
H D EDG D G ??
HG E ??
D E F G H I ??
D E F G H I ??

mgbdZevguc _dlhj
IV$_{AEAD}$ E HD F
<uoh^gufbagZq_gbyZe]hjblfZ=HKL ² ©FZ]fZª\ j_`bf_ 0*0 kh]eZkgh
J ² y\eyxlky
rbnjl_dkl aZrbnjh\Zggh_ ,3OLU l_eh
(QFU\SWHG,3OLU%RG\
I G IG F G E H G ??
E E D D I IFD E D??
E IGFF F F F F F E ??
G I EF HG H HH G ??
FF
bfblh\klZ\dZ kd\hagZy
,QWHJULW\&KHFN9DOXH
Lh]^Z ihke_ rbnjh\Zgby b kd\haghc bfblhaZsblu ,3OLU khh[s_gb_ ijbgbfZ_l ke_^mxs
M' F ??
F DDGEF??
??
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HH ??
E HD F ??
I G IG F G E H G ??
E E D D I IFD E D??
E IGFF F F F F F E ??
G I EF HG H HH G ??
FF??
G E F ??
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$ <uiheg_gb_ ljZgablghc bfblhaZsblu
NeZgZebqbye_ ceyljZgablghc bfblhaZsb\u3OLU khh[s_gb_ bfb_ bagZq_gb_ ihwlhf^mey
khh[s_gby lj_[m_lky \uiheg_gb_ ljZgablghc bfblhaZsblu
lmklv b^_glbnbdZlhj ljZgablgh]h maeZ jZ_g
7UDQVLW,GHQWL¿HU
b ljZgablgZy kbgojhihkuedZ jZ\gZ
7UDQVLW,QLW9DOXH E
$ <uqbke_gb_ ijhba\h^gh]h dexqZ
>ey\uqbke_gb_ ijhba\h^gh]h dexqZ ZjZgablghc bfblhaZsblu Zd_lkZ$_{70\$\&}$gZ\oh^Ze]hjblf=HKL
² ©FZ]fZª \ j_`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ih^Zxlky
dexq dexq h[f_gZ
K'PDVWHU||HHGGFFEEDD ??
I I I I I I I I I I IDIEIFIGIHII

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ijb \uqbke_gbb qZklb dexqZ K

,QW7R9HF_ /DEHO KDF D/ 61 7,91RGH __ k/ __ R/

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FE EG E??

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ijb \uqbke_gbb qZklb dexqZ K

,QW7R9HF_ /DEHO KDF D/ 61 7,91RGH __ k/ __ R/

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ijb \uqbke_gbb qZklb dexqZ K

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ijb \uqbke_gbb qZklb dexqZ K

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<uoh^gu agZq_gb Zef]hjblf=ZHKL ² ©FZ]fZ\ªj_`bf_\ujZ[hldbfblh\klZ\kdb]eZkgh
=HKL ² y\ey_lky khhl_lkl\mxsZy qZklv dexqZ

K GH I I D D

K E G H

K D DDI H D

K EE ID HD

Blh]h\h_ agZq_gb_ dexqZ Zgh

K 70$& GH I I D D E G H ??

D DDI H D EE ID HD

$ LjZgablgZy bfblhaZsblZ

>ey\uiheg_gb ljZgablghc bfblhaZsblu iZd_lZ \oh^Ze]hjblf=HKL ² ©FZ]fZ\ªj_`bf_0*0
kh]eZkgh J ² ih^Zxlky

dexq dexq ljZgablghc bfblhaZsblu iZd_lZ

K 70$& GH I I D D E G H ??

D DDI H D EE ID HD

^hihegbl_ebf bfblhaZsbsZ_f^uZggu_ 3OLU aZ]hZ\hmjh\Zgg,3OLU lagZnq_gibye_c
,QWHJULW\&KHEQ9LOXHUHQQ¿W,QLW9DOXH

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

9HUVLRQ || 7 || &'6 || ([W || '([W61 || '$ **5** ||| .1 || 7.1 || 7LPHVWDPS || 6RXUFH,GHQWL¿HU |
WLRQ,G || HQWLI¿HU || X,RQLHWU9D(QXH \|SWHG || 3,QLWU%JRUG\W\&K7HUFDQ9DOXH || G || HQWL¿HU
7UDQVLW,QLW9DOXH

F ??
F DDGEF??
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HH ??
E HD F ??
I G IG F G E H G ??
E E D D I IFD E D??
E IGFF F F F F F E ??
G I EF HG H HH G ??
FF??
G E F ??
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FE EG E
hldjuluc l_dkl ,3OLU l_eh
QX•O9
mgbdZevguc _dlhj
IV 70$& FE EG E
<uoh^gufbagZq_gbyZe]hjblfZ=HKL ² ©FZ]fZ²\ j_`bf_ 0*0 kh]eZkgh
J ² y\eyxlky
rbnjl_dkl
QX•O9
bfblh\klZ\dZ ljZgablgZy
7UDQVLW,QWHJULW\&KHFN9DOXH E G
Lh]^Z ihke_ ljZgablghc bfblhaZsblu ,3OLU khh[s_gb_ ijbgbfZ_l ke_^mxsbc \b^
M” F ??
F DDGEF??
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HH ??
E HD F ??
I G IG F G E H G ??
E E D D I IFD E D??
E IGFF F F F F F E ??
G I EF HG H HH G ??
FF??
G E F ??
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FE EG E??
E G
$ Khh[s_gb_ M_2
lmklv ,3OLU khh[sihgbe_ MZs__ djbilh]jZnbq_kdhc aZsbl_ bfb^l ke_^mxsbc
M I ??
F DDGEF??
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HH D??
E HD F ??
H D EDG D G ??
HG E ??
D E F G H I ??
D E F G H I ??
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ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

Lh]^Z ihey ,3OLU khhl[s^_gjbZl ke_^mxsb_ agZq_gby
9HUVLRQ
CS
7
D
([W,'
([W61
DAR
5
KN
7.1
7LPHVWDPS F DDGEF
6RXUFH,GHQWL¿HU
'HVWLQDWLRQ,GHQWL¿HU
6HTXHQFH1XPEHU HH D
,QLW9DOXH HD F
3D\ORDG'DWD H D EDG D G ??
HG E ??
D E F G H I ??
D E F G H I
0RGH
7/9
S
5
1H[W+HDGHU
,QWHJULW\&KHFN9DOXH
7UDQVLW,GHQWL¿HU
7UDQVLW,QLW9DOXH
7UDQVLW,QWHJULW\&KHFN9DOXH

B^_glbnbdZ[hjZ3OLU khhl[s_gfb_ agZq_gby ihwlhfdjbilh]jZnbq_kdZy
sblZ khh[s_gby \uihegy_lky kh]eZkgh hibkZgbx djbilh]jZnbq_kdh]h gZ[hjZ 0$*0$ 0*
$ <uiheg_gb_ rbnjh\Zgby b kd\haghc bfblhaZsblu
$ <uqbke_gb_ ijhba\h^gh]h dexqZ
>ey\uqbke_gb_ ijhba\h^gh]h dexqZ gZ\oh^Ze]hjblf=HKL ² ©FZ]fZ\ªj_`bf_\u
jZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ih^Zxlky
dexq dexq h[f_gZ

K PDVWHU DDEEFFGGHHII ??
IHGFED DEFGHI
^Zggu_
ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HF_ /DEHO KDF D/61 ,9 1RGH k/ R/
8
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E HD F ??
HH D??
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ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HF_ /DEHO KDF D/61 ,9 1RGH k/ R/
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E HD F ??
HH D??
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ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HF_ /DEHO$_{KDF}$ D/61 ,9 1RGH k/ R/
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E HD F ??
HH D??
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ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HF_ /DEHO$_{KDF}$ D/61 ,9 1RGH k/ R/
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E HD F ??
HH D??
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<uoh^gufgZq_gb_ef]hjblf=ZHKL ² ©FZ]fZ\ªj_`bf_\ujZ[hldbfblh\klZ\kdb]eZkgh
=HKL ² y\ey_lky khhl_lkl\mxsZy qZklv dexqZ
K E GIGHH HGH
K IEG F I
K_3 E I E GF E G
K I E G FH
Blh]h\h_ agZq_gb_ dexqZ K
K_{AEAD} E GIGHH HGH IEG F I ??
E I E GF E GI E G FH
$ Rbnjh\Zgb_ b kd\hagZy bfblhaZsblZ
>ey\uiheg_gb bynjh\Zgb b kd\hagh bfblhaZsblZd_lZ ZZ\oh^Ze]hjblf=HKL ² ©FZ]
fZª \ j_`bf_ 0*0 kh]eZkgh J ² ih^Zxlky
dexq dexq rbnjh\Zgby b kd\haghc bfblhaZsblu iZd_lZ
K_{AEAD} E GIGHH HGH IEG F I ??
E I E GF E GI E G FH
^hihegbl_evgu_ bfblhaZsbsZ_fu_ ^Zggu_ ,3OLU aZ]heh\hd
9HUVLRQ || &6 || ([W || '([W61 || '$ 5 ||||.1 || || 7LPHVWDPSU||FH,G||H'DWD&RQW
WLRQ,G||H&RQWHQW&HUWLI||LFDWH1||X,RQHWW9DOXH
??
F DDGEF??
??
??
HH D ??
E HD F
hldjuluc l_dkl ,3OLU l_eh
3D\ORDG||'DWD&RQW||9||S|| 5 ||1H[W+HDGHU
H D EDG D G ??
HG E ??
D E F G H I ??
D E F G H I ??

mgbdZevguc _dlhj
IV_{AEAD} E HD F

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

<uoh^gufbagZq_gbyZe]hjblfZ=HKL ² ©FZ]fZª\ j_`bf_ 0*0 kh]eZkgh
J ² y\eyxlky
rbnjl_dkl aZrbnjh\Zggh_ ,3OLU l_eh
(QFU\SWHG,3OLU%RG\
G IIFD G IFEFDH F I HI F ??
F E HIE FEHGG I E F ??
I E I D G G F HI ??
G D D DIE HIII FF DEH D ??
D

bfblh\klZ\dZ kd\hagZy
,QWHJULW\&KHFNHDOXH

Lh]^Z ihke_ rbnjh\Zgby b kd\haghc bfblhaZsblu ,3OLU khh[s_gb_ ijbgbfZ_l ke_
M' I ??
F DDGEF??
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HH D??
E HD F ??
G IIFD G IFEFDH F I HI F ??
F E HIE FEHGG I E F ??
I E I D G G F HI ??
G D D DIE HIII FF DEH D ??
D??
HIHH D??
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$ <uiheg_gb_ ljZgablghc bfblhaZsblu
NeZ jZebqbye_ ^eyjZgablghc bfblhaZsblu ,3OLU khh[s_gb_ ijbgbfZ_l ke_ agZq_gb_ 7b_ ihwlhfme^y
khh[s_gby lj_[m_lky \uiheg_gb_ ljZgablghc bfblhaZsblu
lmklv b^_glbnbdZlhj ljZgablgh]h maeZ jZ_g
7UDQVLW,GHQWL¿HU
b ljZgablgZy kbgojhihkuedZ jZ\gZ
7UDQVLW,QLW9DOXH E
$ <uqbke_gb_ ijhba\h^gh]h dexqZ
>ey\uqbke_gb_ ijhba\h^gh]h dexqZ jZgablghc bfblhaZsblu Zd_lZ 70$& gZ\oh^Ze]hjblfm=HKL
² ©FZ]fZª \j_`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ih^Zxlky
dexq dexq h[f_gZ
K' PDVWHU IIHHGGFFEEDD ??
I I I I I I I I I I IDIEIFIGIHII
^Zggu_
ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HE_ /DEHO KDF D/ 61 7,91RGH __ k/ __ R/
8
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G ??
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FE EG E??
HH D??
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ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HE_ /DEHO KDF D/ 61 7,91RGH __ k/ __ R/
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G ??
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FE EG E??

HH D??
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ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HF__ /DEHO KDF D/ 61 7,91RGH __ k/ __ R/
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FE EG E??
HH D??
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ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HF__ /DEHO KDF D/ 61 7,91RGH __ k/ __ R/
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FE EG E??
HH D??
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ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

hldjuluc l_dkl ,3OLU l_eh

QXOO

mgbdZevguc _dlhj

IV 70$& FE EG E

<uoh^gufbagZq_gbyZbe]hjblfZ=HKL ² ©FZ]fZª\ j_`bf_ 0*0 kh]eZkgh

J ² y\eyxlky

rbnjl_dkl

QXOO

bfblh\klZ\dZ ljZgablgZy

7UDQVLW,QWHJULW\&KHFN9DOXH EFGI EI

Lh]^Z ihke_ ljZgablghc bfblhaZsblu ,3OLU khh[s_gb_ ijbgbfZ_l ke_^mxsbc \b^

M" l ??

F DDGEF??

??

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HH D??

E HD F ??

G IIFD G IFEFDH F I HI F ??

F E HIE FEHGG I E F ??

I E I D G G F HI ??

G D D DIE HIII FF DEH D ??

D??

HIHH D??

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FE EG E??

EFGI EI

$ Djbilh]jZnbq_kdbc gZ[hj .8=1 &75 &0$&

$ Khh[s_gb M_3

Imklv ,3OLU khh[s_gb_ ^ZZs_ djbilh]jZnbq_kdhc aZsbl_ bf\b^l ke_^mxsbc

M_3 F ??

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E HD F ??

H D EDG D G ??

HG E ??

D E F G H I ??

D E F G H I ??

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Lh]^Z ihey ,3OLU khh[s_gb_ ^ZZl ke_^mxsb_ agZq_gby

9HUVLRQ

CS

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7.1

7LPHVWDPS F DDGEF

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

6RXUFH,GHQWL¿HU
'HVWLQDWLRQ,GHQWL¿HU
6HTXHQFH1XPEHHU
,QLW9DOXH HD F
3D\ORDG'DWD H D EDG D G ??
HG E ??
D E F G H I ??
D E F G H I
0RGH
7/9
S
5
1H[W+HDGHU
,QWHJULW\&KHFN9DOXH
7UDQVLW,GHQWL¿HU
7UDQVLW,QLW9DOXH
7UDQVLW,QWHJULW\&KHFN9DOXH
B^_glbnbdZ djhjilhgZ[h\jZ3OLU khhl[s_gfb_b_lagZq_gb5_ ihwlhfdhjbilh]jZnbq_kdZy
sblZ khh[s_gby \uihegy_lky kh]eZkgh hibkZgbx djbilh]jZnbq_kdh]h gZ[hjZ .8=1 &75 &0$
$ <uiheg_gb_ rbnjh\Zgby b kd\haghc bfblhaZsblu
$ <uqbke_gb_ ijhba\h^guo dexq_c
>ey\uqbke_gb_ ijhba\h^guo dexq_Kc bK gZ\oh^Ze]hjblfmHKL ² ©Dmag_ijd ª
`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ih^Zxlky
dexq dexq h[f_gZ
K DDEEFFGGHHII ??
IHGFED DEFGHI
^Zggu_
ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HF_ /DEHO__D/61__,9 1RGH __ k/ __ R/
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H G ??
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E HD F ??
HH ??
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ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HF_ /DEHO__D/61__,9 1RGH __ k/ __ R/
??
H G ??
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E HD F ??
HH ??
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ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HF_ /DEHO__D/61__,9 1RGH __ k/ __ R/
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H G ??
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E HD F ??
HH ??
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ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HF__ /DEHO$_{KDF}$__D/61__,9 1RGH __ k/ __ R/
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H G ??
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E HD F ??
HH ??
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<uoh^gu afgZq_gb Z_ef]hjblf=ZHKL ² ©Dmag_qb_d`bf_\ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² y\ey_lky khhl_lkl\mxsZy qZklv dexqZ

K E G D D H DF E G
K E D EGD H I G H
K_3 IF H IG E E G
K G D H EFFDF EG H H
Blh]h\u_ agZq_gby dexb K_{ENC} c K_{MAC} kjZ\gu
K_{ENC} E G D D H DF E G??
E D EGD H I G H
K_{MAC} IF H IG E E G??
G D H EFFDF EG H H
$ Rbnjh\Zgb_ b kd\hagZy bfblhaZsblZ
>ey\uiheg_gby bbynjh\Zgby iZd_lZ ZZ\oh^Ze]hjblf=mHKL ² ©Dmag_qb_d`bf_]Zffbjh \Zgby kh]eZkgh =HKL ² ih^Zxlky
dexq dexq rbnjh\Zgby iZd_lZ
K_{ENC} E G D D H DF E G??
E D EGD H I G H
hldjuluc l_dkl ,3OLU l_eh
3D\ORDG||'DWD7||9||S||5 ||1H[W+HDGHU
H D EDG D G ??
HG E ??
D E F G H I ??
D E F G H I ??

kbgojhihkuedZ
IV_{ENC} E HD F
<uoh^gu afgZq_gb Z_ef]hjblf=ZHKL ² ©Dmag_qb_d`bf_]Zffbjh\Zgby kh]eZkgh =HKL ² y\ey_lky
rbnjl_dkl aZrbnjh\Zggh_ ,3OLU l_eh
(QFU\SWHG,3OLU%RG\
F H D F FD GF ??
HH I F D D H E I ??
H IF E HD FH G ??
F F G F F D G I ??
G E
>ey\uiheg_gby bbynjh\Zgby iZd_lZ ZZ\oh^Ze]hjblf=mHKL ² ©Dmag_qb_d`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ih^Zxlky
dexq dexq kd\haghc bfblhaZsblu iZd_lZ
K_{MAC} IF H IG E E G??
G D H EFFDF EG H H
^Zggu_ ,3OLU aZ]heh\hd aZrbnjh\Zggh_ ,3OLU l_eh
9HUVLRQ||&RG||([W||'([W61||'$7||.1||7LPHVWDPS||&RXQWHU||&RQWHQW||&RQW
WLRQ,G||&RQWHQW&RQWHQW&RQWHQW&RQWHQW&RQWHQW&RQWHQW&RQWHQW&RQW
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F DDGEF??
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ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

HH ??
E HD F ??
F H D F FD GF ??
HH I F D D H E I ??
H IF E HD FH G ??
F F G F F D G I ??
G E

<uoh^gu afgZq_gb Z_ef]hjblf=ZHKL ² ©Dmag_qbd^a bf_\ujZ[hldb bfblh\klZ\db kh]eZk
gh =HKL ² y\ey_lky
bfblh\klZ\dZ kd\hagZy

,QWHJULW\&KHFN9DOXH I H G

Lh]^Z ihke_ rbnjh\Zgby b kd\haghc bfblhaZsblu ,3OLU khh[s_gb_ ijbgbfZ_l ke_^mxs
M'3 F ??
F DDGEF??
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HH ??
E HD F ??
F H D F FD GF ??
HH I F D D H E I ??
H IF E HD FH G ??
F F G F F D G I ??
G E??
HH HH I H G??
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$ <uiheg_gb_ ljZgablghc bfblhaZsblu
NeZgjZebqbye_^ey ljZgablghc bfblhaZsb\u3OLU khh[s_gb_ ijbgbfZ_l ke_^mxs M3 bfb_hagZq_gb_ ihwlhf^ey
khh[s_gby lj_[m_lky \uiheg_gb_ ljZgablghc bfblhaZsblu
lmklv b^_glbnbdZlhj ljZgablgh]h maeZ jZ_g

7UDQVLW,GHQWL¿HU
b ljZgablgZy kbgojhihkuedZ jZ\gZ

7UDQVLW,QLW9DOXH E

$ <uqbke_gb_ ijhba\h^gh]h dexqZ
>ey\uqbke_gb_ ijhba\h^gh]h dexqZ ljZgablgh bfblhaZsblu lZd_lkZ 70$& gZ\oh^Ze]hjblf=HKL
² ©Dmag_qbd^a \j_`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ih^Zxlky
dexq dexq h[f_gZ

K'PDVWHU ||HHGGFFEEDD ??
I I I I I I I I I IDIEIFIGIHII
^Zggu_
ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HF__ /DEHO KDF D/ 61 7,91RGH __ k/ __ R/
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G ??
??
FE EG E??
HH ??
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ijb \uqbke_gbb qZklb dexqZ K
,QW7R9HF__ /DEHO KDF D/ 61 7,91RGH __ k/ __ R/
??
G ??
??
FE EG E??
HH ??

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

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<u o h ^ g u afgZq_gb Z_ef]hjblf=ZHKL ² ©Dmag_qb_d`abf_\ujZ[hldbfblh\klZ\db kh]eZk
gh =HKL ² y\ey_lky khhl_lkl\mxsZy qZklv dexqZ

K HD H G IGF I I GID E FE

K E FFD F I H

Blh]h\h_ agZq_gb_ dexqZ qZgh

$K_{70\$\&}$ HD H G IGF I I GID E FE??

E FFD F I H

$ LjZgablgZy bfblhaZsblZ

>ey\uiheg_gb ljZgablghc bfblhaZsblu iZ\oh^Ze]hjblf=ZHKL ² ©Dmag_qb_d`abf_\u
jZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ih^Zxlky

dexq dexq ljZgablghc bfblhaZsblu iZd_lZ

$K_{70\$\&}$ HD H G IGF I I GID E FE??

E FFD F I H

^Zggu_3OLU aZ]ahhjh\Zggh3OLU agZq_gb ey_cQWHJULW\&KHFN9DOXH 7UDQVLW,QLW9DOXH
7UDQVLW,QLW9DOXH

9HUVLRQ || 7 || &'6 || ||([W || ,'([W61 || '$5 || ||.1 || 7.1 || 7LPHVWDPS || 6RXUFH,GHQWL
WLRQ,G || HQWLX&HQFH1|X,RELHW9DOXH || (QFU||SWHG,3DOW&KHFN9DOXH||G||HQWL
7UDQVLW,QLW9DOXH

F ??

F DDGEF??

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HH ??

E HD F ??

F H D F FD GF ??

HH I F D D H E I ??

H IF E HD FH G ??

F F G F F D G I ??

G E??

HH HH I H G??

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FE EG E

<u o h ^ g u afgZq_gb Z_ef]hjblf=ZHKL ² ©Dmag_qb_d`abf_\ujZ[hldbfblh\klZ\db kh]eZk
gh =HKL ² y\ey_lky

bfblh\klZ\dZ ljZgablgZy

7UDQVLW,QWHJULW\&KHFN9DOXH IEH EFI FE

Lh]^Z ihke_ ljZgablghc bfblhaZsblu ,3OLU khh[s_gb_ ijbgbfZ_l ke_^mxsbc \b^

M"$_3$ F ??

F DDGEF??

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HH ??

E HD F ??

F H D F FD GF ??

HH I F D D H E I ??

H IF E HD FH G ??

F F G F F D G I ??

G E??

HH HH I H G??

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FE EG E??

IEH EFI FE

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

$ Khh[s_gby M_4
lmklv ,3OLU khh[s_gby ihgbe_ MZs__ djbilh]jZnbq_kdhc aZsbl_ bfb^l ke_^mxsbc
M l ??
F DDGEF??
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HH E??
E HD F ??
H D EDG D G ??
HG E ??
D E F G H I ??
D E F G H I ??
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Lh]^Z ihey ,3OLU khh[s_gby Mhhk[s_gby Zl ke_^mxsb_ agZq_gby
9HUVLRQ
CS
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([W61
DAR
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KN
7.1
7LPHVWDPS F DDGEF
6RXUFH,GHQWL¿HU
'HVWLQDWLRQ,GHQWL¿HU
6HTXHQFH1XPEHU HH E
,QLW9DOXH HD F
3D\ORDG'DWD H D EDG D G ??
HG E ??
D E F G H I ??
D E F G H I
0RGH
7/9
S
5
1H[W+HDGHU
,QWHJULW\&KHFN9DOXH
7UDQVLW,GHQWL¿HU
7UDQVLW,QLW9DOXH
7UDQVLW,QWHJULW\&KHFN9DOXH
B^_glbnbdZ djhjilhgZ[h\jZ 3OLU khh[s_gby bfb_b_ agZq_gby CS_ ihwlhfdhjbilh]jZnbq_kdhc aZZy
sblZ khh[s_gby \uihegy_lky kh]eZkgh hibkZgbx djbilh]jZnbq_kdh]h gZ[hjZ .8=1 &75 &0$
$ <uiheg_gb_ rbnjh\Zgby b kd\haghc bfblhaZsblu
$ <uqbke_gb_ ijhba\h^guo dexq_c
>ey\uqbke_gb_ ijhba\h^guo dexq_c K_{ENC} bK $_{MAC}$ gZ\oh^Ze]hjblf=HKL ² ©Dmag_lb_d ª
`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ih^Zxlky
dexq dexq h[f_gZ
K $_{PDVWHU}$ DDEEFFGGHHII ??
IHGFED DEFGHI

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

^Zggu_

ijb \uqbke_gbb qZklb dexqZ K

,QW7R9HF_ /DEHO$_{KDF}$ D/61,9 1RGH k/ R/

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H G ??

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E HD F ??

HH E??

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ijb \uqbke_gbb qZklb dexqZ K

,QW7R9HF_ /DEHO$_{KDF}$ D/61,9 1RGH k/ R/

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ijb \uqbke_gbb qZklb dexqZ K

,QW7R9HF_ /DEHO$_{KDF}$ D/61,9 1RGH k/ R/

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<uoh^gu agZq_gby Zef]hjblf=HKL ² ©Dmag_ub_d`abf_\ujZ[hldbfblh\klZ\kdb]eZk gh =HKL ² y\ey_lky khhl_lkl\mxsZy qZklv dexqZ

K E D F GD E I

K IE H IH D E D D

K_3 D FD GI E IFE ID E ID H

K G G E H H FDG

Blh]h\u_ agZq_gby dexb K_{ENC} c K_{MAC} jZ\gu

K_{ENC} E D F GD E I ??

IE H IH D E D D

K_{MAC} D FD GI E IFE ID E ID H ??

G G E H H FDG

$ Rbnjh\Zgb_ b kd\hagZy bfblhaZsblZ

>ey\uiheg_gby rbnjh\Zgby iZd_lZ ZZ\oh^Ze]hjblf=HKL ² ©Dmag_ub_d`abf_]Zffbjh \Zgby kh]eZkgh =HKL ² ih^Zxlky

dexq dexq rbnjh\Zgby iZd_lZ

K_{ENC} E D F GD E I ??

IE H IH D E D D

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

hldjuluc l_dkl ,3OLU l_eh

3D\ORDQ||'DWD7||9||S||5 ||1H[W+HDGHU

H D EDG D G ??

HG E ??

D E F G H I ??

D E F G H I ??

kbgojhihkuedZ

IV_{ENC} E HD F

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rbnjl_dkl aZrbnjh\Zggh_ ,3OLU l_eh

(QFU\SWHG,3OLU%RG\

EFEFHHG D FF D F ED ??

F DGF FE EI E H E H HG??

D G G DFI F EE D E IHGF ??

H IID G F DD E F H H ??

HHE

>ey\uiheg_gbd\haghc bfblhaZsbluZd_lZZ\oh^Ze]hjblf=HKL ² ©Dmag_q\bd_a`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ih^Zxlky

dexq dexq kd\haghc bfblhaZsblu iZd_lZ

K_{MAC} D FD GI E IFE ID E ID H ??

G G E H H FDG

^Zggu_ ,3OLU aZ]heh\hd aZrbnjh\Zggh_ ,3OLU l_eh

9HUVLRQ || &&||([W||'([W61 || '$5 |||.1 || || 7LPHVWDPSU||FH,G||H'DWD&RQW||HQW&RQW

WLRQ,G||H&RQW&RQW||H1||X,RQHW&RQW&RQW||$WHG,3OLU%RG\

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F DDGEF??

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HH E??

E HD F ??

EFEFHHG D FF D F ED ??

F DGF FE EI E H E H HG??

D G G DFI F EE D E IHGF ??

H IID G F DD E F H H ??

HHE

<uoh^gu afgZq_gbZ_e]hjblf=HKL ² ©Dmag_q\bd_a`bf_\ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² y\ey_lky

bfblh\klZ\dZ kd\hagZy

,QWHJULW\&KHFN9DOXH

Lh]^Z ihke_rbnjh\Zgby b kd\haghc bfblhaZsblu ,3OLU khh[s_gb_ ijbgbfZ_l ke_^mxs

M' l ??

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HH E??

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ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

$ <uiheg_gb_ ljZgablghc bfblhaZsblu

NeZ[ljZebqbhye_ceylj Zgablghc bfblhaZsbl\u3OLU khhl[s_gfb_b_agZq_g7b_ ihwlhfmey khh[s_gby lj_[m_lky \uiheg_gb_ ljZgablghc bfblhaZsblu

lmklv b^_glbnbdZlhj ljZgablgh]h maeZ jZ_g

7UDQVLW,GHQWL¿HU

b ljZgablgZy kbgojhihkuedZ jZ\gZ

7UDQVLW,QLW9DOXH E

$ <uqbke_gb_ ijhba\h^gh]h dexqZ

>ey\uqbke_gb ijhba\h^gh]h dexqZ ljZgablghc bfblhaZsblu iZd_lZ K70$& gZ\oh^Ze]hjblfm =HKL ² ©Dmag_qbd^a \ j_`bf_ \ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ²

dexq dexq h[f_gZ

K'PDVWHU lJHHGGFFEEDD ??

I I I I I I I I I I IDIEIFIGIHII

^Zggu_

ijb \uqbke_gbb qZklb dexqZ K

,QW7R9HF__ /DEHO KDF D/ 61 7,91RGH __ k/ __ R/

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G ??

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FE EG E??

HH E??

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ijb \uqbke_gbb qZklb dexqZ K

,QW7R9HF__ /DEHO KDF D/ 61 7,91RGH __ k/ __ R/

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FE EG E??

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K I FDD H I IH D I

K D IGG D GHH IH G H H

Blh]h\h_ agZq_gb_ dexqZ 70$& ZgkH

K70$& I FDD H I IH D I ??

D IGG D GHH IH G H H

$ LjZgablgZy bfblhaZsblZ

>ey\uiheg_gb ljZgablghc bfblhaZsblu iZ\oh^Ze]hjblf=HKL ² ©Dmag_qbd^a bf_\ujZ[hldb bfblh\klZ\db kh]eZkgh =HKL ² ih^Zxlky

dexq dexq ljZgablghc bfblhaZsblu iZd_lZ

K70$& I FDD H I IH D I ??

D IGG D GHH IH G H H

^Zggu_3OLU aZ]aZ h\ jh\Zgg3OLU agZq_gb ye_cQWHJULW\&7HUPLQDO&HQW

7UDQVLW,QLW9DOXH

9HUVLRQ || 7|&'6||([W||,'([W61 || '$5 |||.1 || 7.1 || 7LPHVWDPS || 6RXUFH,GHQWL

WLRQ,G||HQWLX&HQFH1X,PEHW9DOXH || (QFU||SWHG3DW&KHFN9DOXH||HQWL

7UDQVLW,QLW9DOXH

I ??

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HH E??

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

E HD F ??
EFEFHHG D FF D F ED ??
F DGF FE EI E H E H HG??
D G G DFI F EE D E IHGF ??
H IID G F DD E F H H ??
HHE ??
D FH ??
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FE EG E

<u o h ^ g u a f g Z q _ g b Z _ e f] h j b l f = Z H K L ² © D m a g _ q b d ` a b f _ \ u j Z [h l d b f b l h \ k l Z \ d b] e Z k
g h = H K L ² y \ e y _ l k y
b f b l h \ k l Z \ d Z l j Z g a b l g Z y

7 U D Q V L W , Q W H J U L W \ & K H F N 9 D O X H G I G E H

L h] ^ Z i h k e _ l j Z g a b l g h c b f b l h a Z s b l u , 3 O L U k h h [s _ g b _ i j b g b f Z _ l k e _ ^ m x s b c \ b ^
M" I ??
F DDGEF??
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HH E??
E HD F ??
EFEFHHG D FF D F ED ??
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ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

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fZjrjmlbaZpby 931 ,3 iZd_l lmgg_ev

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ТЕХНИЧЕСКОМУ РЕГУЛИРОВАНИЮ И МЕТРОЛОГИИ

J_^Zdlhj E K Abfbeh\Z
L_ogbq_kdbc j_^Zdlhj B ? Q_j_idh\Z
Dhjj_dlhj H < EZaZj_\Z
Dhfivxl_jgZy _jkldZ E : Djm]h\hc

K^Zgh \ gZ[hj Ih^ibkZgh \ i_qZlv 1 8 N=Zjgdlmjiz :jbZe
Mke i_q e Mq ba^ e
Ih^]hlh\e_gh gZ hkgh_ we_dljhgghc _jkbb ij_^hklZ\e_gghc jZajZ[hlqbdhf klZg^

Kha^Zgh \ _^bgbqghf bkiheg_gbb \h N=MI ©KL:G>:JLBGNHJFª
^ey dhfie_dlh\Zgby N_^_jZevgh]h bgnhjfZpbhggh]h nhg^Z klZg^Zjlh\ Fhkd\Z GZobf
ZZZ JRVWLQIR UX LQIR#JRVWLQIR UX